Joseph Ineza Karangwa

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SUMMARY

Data Analyst and AI/ML Engineer with experience building monitoring dashboards, clinical analytics tools, and digital health systems. Skilled in SQL-based data modeling, KPI tracking, and translating operational and product data into structured reporting for program and clinical teams. Experience supporting health-focused platforms with analytics, safety logging, and performance monitoring in resource-constrained contexts.

Education

Master of Science in Engineering Artificial Intelligence **August 2024 - May 2026**
Carnegie Mellon University
Specializations: Applied Machine Learning

Bachelor of Science with Honors in Software Engineering **December 2020 - November 2024**
Adventist University of Central Africa
Specializations: Software Engineering

PROFESSIONAL EXPERIENCE

AI/ML Integration Consultant **December 2025 – Present**
Mindora Health, Kigali - Rwanda

- Facilitated intern interviews aligned with chatbot performance metrics and dashboard goals.
- Provide AI integration and analytics support for a digital mental health platform serving Rwandan youth.
- Design structured logging frameworks for emotion detection and interaction metadata to enable monitoring of usage trends and safety signals.
- Support development of internal evaluation dashboards to track engagement patterns and system behavior.
- Collaborate with product and clinical stakeholders to translate system outputs into measurable indicators for continuous improvement.

AI/ML Integration Lead - Capstone **August 2025 – December 2025**
Mindora Health, Kigali - Rwanda

- Engineered a context-aware LLM-based emotion classification system replacing keyword-driven logic.
- Built longitudinal emotion-logging models linked to user and conversation IDs to support reporting and review.
- Implemented crisis detection and safety layers aligned with responsible AI deployment standards.

AI and LLM Engineer Intern **June 2025 - August 2025**
Upanzi Network at CMU-Africa, Kigali - Rwanda

- Architected a Retrieval-Augmented Generation (RAG) study assistant using structured ETL pipelines and vector embeddings.
- Designed data-processing workflows to normalize heterogeneous educational materials into analytics-ready formats.
- Produced evaluation summaries and reporting insights for stakeholder review.
- Recognition: Featured by the Global Cyber Alliance for contributions to democratizing cybersecurity education through AI.
- ICT Training Coordination: Coordinated the 2025 ICT Teachers Training curriculum in partnership with World Vision Rwanda, managing a 3-day program for 30 educators from schools supporting Nyabiheke, Kigeme, and Mugomwa refugee camps.
- Facilitated a specialized technical workshop on Geographic Information Systems (GIS), providing ICT teachers with practical tools for digital mapping and spatial data in the ICT Teachers Training.

Software Engineer Intern **January 2024 - May 2024**
Hview Tech Group, Kigali - Rwanda.

- Collaborating on development of Sawa Telematics Product to address transportation challenges across Africa.
- Conducting front-end development using Vue.js, focusing on responsive design to enhance user experience and delivering insightful dashboards.
- Reported weekly and monthly on achievements, challenges faced, and the way forward to enhance team performance and operational efficiency.

Order Operations Manager **June 2020 - December 2020**
Huzalabs, Kigali - Rwanda.

- Managed marketing efforts and analyzed customer feedback data to refine outreach strategies and improve lead engagement.
- Managed the product catalog and coordinated order-processing workflows, improving delivery efficiency and pricing.

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RELEVANT PROJECTS

Clinical Operations & Patient Experience Dashboard **Dashboard View**

- Modeled 53,000+ consultation records in PostgreSQL to track operational KPIs including referral rates, clinician workload, and service utilization.
- Integrated synthetic patient-experience signals (NPS, complaint categories, comment analysis) to support quality improvement review.
- Developed interactive dashboards enabling longitudinal operational and service-quality monitoring.

Rwanda Child Nutrition MEL Dashboard **Dashboard View**

- Built an end-to-end monitoring dashboard from UNICEF JME survey data using PostgreSQL and Streamlit.
- Modeled analytics views for KPI snapshots, trend analysis, method comparison, and stratifier equity review.
- Delivered year-over-year delta indicators and exportable reports for stakeholder reporting.
- Consolidated 10 years of nutrition indicators into a decision-ready monitoring system.

RAG System for Cancer Myth Detection and Fact-Checking **View on Portfolio**

- Architected an end-to-end RAG system for evidence-grounded cancer myth detection using structured medical reference documents.
- Implemented modular retrievers and generators (OpenAI embeddings, SentenceTransformers, FAISS, GPT-4o-mini, FLAN-T5) to enable systematic performance comparison.
- Built an evaluation pipeline with batch prediction, metric aggregation, statistical analysis, and visualization to assess factual grounding and retrieval quality.
- Improved reliability of health misinformation responses by enforcing evidence-backed answer generation.

Life Prognosis Management Tool **View on Portfolio**

- Implemented patient lifecycle workflows and prognosis estimation logic from structured clinical inputs.
- Developed administrative reporting modules for lifespan-related statistics and data export.
- Strengthened structured data capture for improved reporting consistency.

TECHNICAL SKILLS

- SQL (PostgreSQL)
- SQL views
- Data modeling
- ETL pipelines
- Python (Pandas, NumPy)
- Java
- Streamlit
- PostgreSQL (Neon)
- Supabase
- Git
- Docker
- RAG system

CERTIFICATIONS

- Social & Behavioral Research - Basic/Refresher (Oct 2025) from Collaborative Institutional Training Initiative (CITI).
- AI Augmented Professional Development Skills in the Digital Age (May 2024) from ALX AFRICA | AI Career Essentials.
- NDG Linux Essentials (Nov 2023) from Cisco Networking Academy.
- AWS Machine Learning Foundations (Nov 2022) from Udacity.
- Artificial Intelligence in Natural Language Processing (Nov 2021) from Huzalabs, Rwanda.

AWARDS & ACHIEVEMENTS

- Mastercard Foundation Scholar, part of a network committed to socio-economic transformation in Africa.
- Finalist in the Higher Learning Institutions Exhibitions and STEM Competitions 2023.

LANGUAGES

- Kinyarwanda - Native
- English – Fluent
- French - Beginner

EXTRACURRICULAR ACTIVITIES AND VOLUNTEERING

- Debate Coaching
- Rwanda Traditional Dancing (Amaraba)
- K-12 Engineering outreach
- Youth and Teenagers Experiential Learning

INTERESTS

- Digital health
- Inclusive Technology
- Health, EdTech and Social Tech analytics
- Monitoring & Evaluation
- Responsible AI